Circuits I

Current & Charge

Current is the rate of flow of charge through a section of a conductor, or

$$i(t) = \frac{dq(t)}{dt}$$

and hence the charge can be found by

$$q(t) = \int_{t_0}^{t} i(t)\ dt + q(t_0)$$

Currents can also have a direction given a reference direction, where current flowing in the same direction is positive, and vice versa for current flowing opposite to the reference direction

Current can also be direct; staying at a constant value, or alternating.

Voltages

The energy transferred per unit of charge, either released or absorbed. It can also have a reference direction set by setting one end as positive and another as negative (which represents which side has a higher potential than the other)

we can represent voltages as v_{ab}, representing the voltage measured from a to b with a as the positive side.

Hence,

$$v_{ab} = -v_{ba}$$

Power & Energy

Power across a component is found by

$$p(t) = i(t)v(t)$$

where the polarity of the voltage and current must be taken into account, where a positive power represents power being absorbed by the element. A negative power represents power being supplied by the element.

The sign of p can be determined if the reference current flows in the same direction as the voltage. If the two flow together, p is positive, otherwise p is negative.

In other words, when the current and voltage conventions do not flow together, we can use the formula

$$p = -vi$$

Then, the energy, or work done can be found as

$$w = \int_{t_1}^{t_2} p(t)\ dt$$

where a positive w means energy is put out into the system, or the component does work.

KCL & KVL

KCL states that the algebraic sum of currents at a node is 0, that is

$$\sum i_n = 0$$

or that the sum of currents in is equal to the sum of currents out

This also means that for a series arrangement of components,

$$i_1 = i_2 = ... = i_n$$

KVL states that a loop has the algebraic sum of voltages as 0, where a loop going across an element from positive to negative has a positive sign, and vice versa.

$$\sum v_n = 0$$

Hence for a series loop, the power supplied and used must equal 0, or

$$\sum v_n i_n = 0$$

as a loop will contain components that supply power and components that absorb power, and the total power supplied will be 0 by the conservation of energy.

For two parallel branches,

$$v_1 = v_2 = ... = v_n$$

Circuit Elements

Conductors

An ideal wire; has 0 resistance and 0 voltage across.

Connecting a power source at the ends of a conductor gives a short circuit.

Absence of conductor gives a open circuit.

Resistors

limits flow of electrons in circuit, by ohms law,

$$v = iR \Rightarrow R = \frac{v}{i}$$

where polarity of v and i against reference must be considered. (current flowing opposite to voltage convention gives v= -iR)

p

Resistance is determined by the inherent properties of the resistor, by

$$R = \frac{\rho L}{A}$$

The power dissipated by a resistor can be determined by

$$p = vi = i^2 R = \frac{v^2}{R}$$

where the energy is dissipated as heat.

Voltage & Current Sources

An ideal (independent) voltage source is drawn as a circle with its polarity indicated, and has 0 internal resistance.

Actual voltages sources, or practical voltage sources, have a non zero but small amount of resistance, and can be modelled as a voltage source with a resistor in series.

Voltage sources can also be independent or dependent. Independent voltages have the voltage level constant, or oscillate periodically based on time and are not affected by other components. Dependent voltage sources are indicated by a diamond, and have their voltage level dependent on the voltage/current of a component, aka the voltage is a function of other voltages or currents in the cirucit.

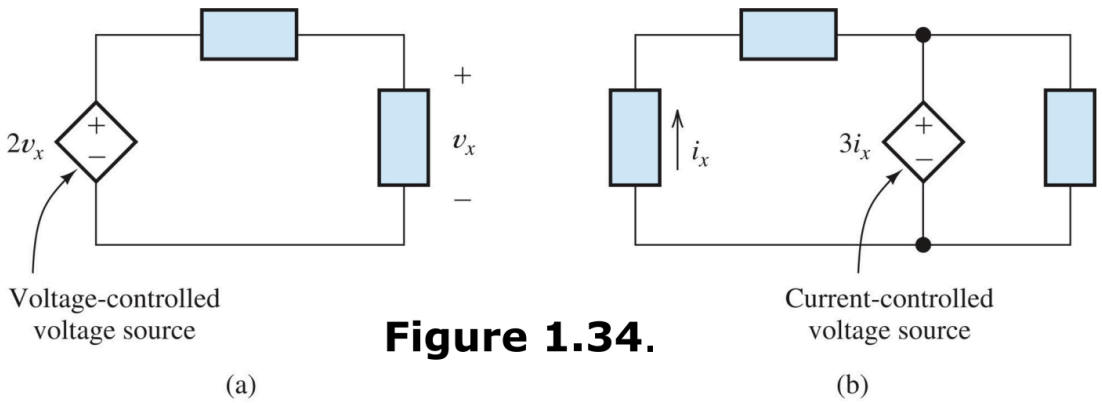


Figure 1.34.

this is cool but idk why you would want this

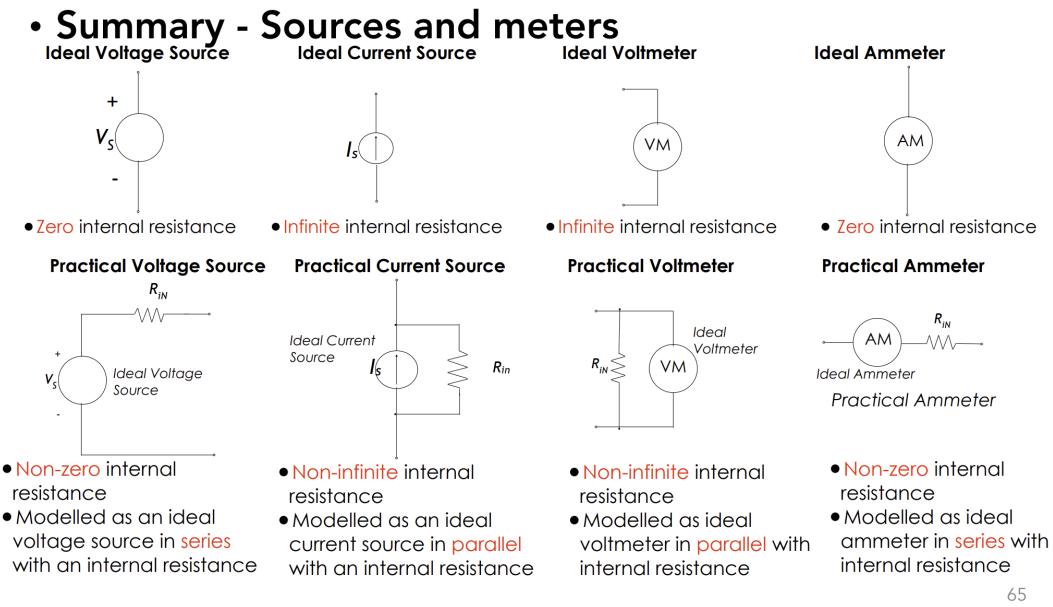
An ideal independent current source is drawn as a circle with an arrow indicating direction, and has infinite resistance, to prevent current from leaking out of the source. However, actual current sources have some finite resistance to them, and are modelled as an ideal current source with a resistor in parallel as an extra branch that allows current to flow, dropping the actual current supplied.

Again, a dependent current source is drawn as a diamond, and can be dependent on either voltage or current. The convention of the read voltage/source must be considered when finding the current supplied.

Ammeters and Voltmeters

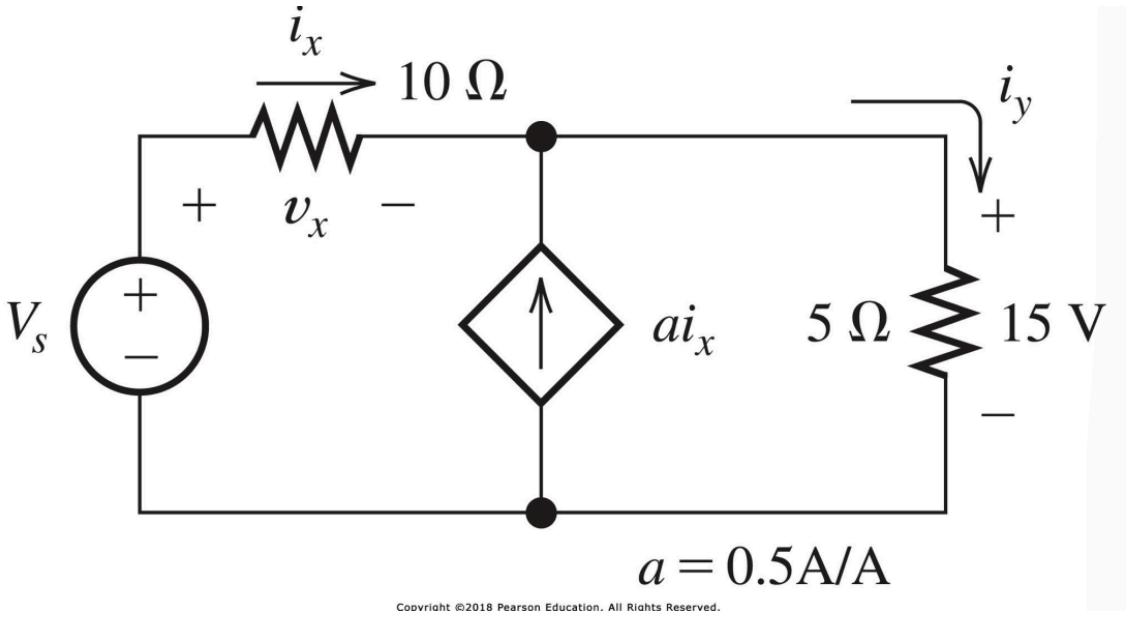
An ideal ammeter measures the current through a branch, and has 0 resistance. In reality, some non zero resistance will exist and is modelled with a resistor in series.

Likewise, an ideal voltameter measures the current across a branch, and has infinite resistance. In reality, some finite resistance is present and is modelled as a resistor in parallel.



Circuit Analysis

Determine V_s on the circuit shown:



$i_y = \frac{15}{5} = 3A$

By KCL, on the top middle node, $i_x + ai_x = i_y$

$1.5i_x = 3$

$i_x = \frac{3}{1.5} = 2A$

$\therefore v_x = i_x R = (2)(10) = 20V$

By KVL on the outermost loop, $v_x + 15 - V_s = 0$

$V_s = v_x + 15$

$V_s = 20 + 15$

$V_s = 35V$

main idea is to work inside out to get desired value